Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/06/2017
PI Name:	Delp, Michael Ph.D.		
Project Title:	Effects of Microgravity on Cerebral Arterial, Venous, and Lymphatic Function: Implications for Elevated Intracranial Pressure		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	12/01/2015	**End Date:**	11/30/2019
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	5	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	ISS Rodent Research-9 NOTE: Extended to 11/30/2019 per NSSC information (Ed., 12/11/18)		
Key Personnel Changes/Previous PI:	March 2017: No changes.		
COI Name (Institution):	Zawieja, David Ph.D., M.D. (Texas A&M University)		
Grant/Contract No.:	NNX16AC28G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Effects of Microgravity on Cerebral Arterial, Venous, and Lymphatic Function: Implications for Elevated Intracranial Pressure," grant #NNX13AN33G, due to PI move to Florida State University. Approximately 29% of astronauts on short-term (~2 wk) space shuttle flights and 60% on long-duration (~6 mo) missions to the International Space Station (ISS) are reported to have experienced some impairment in distant or near visual acuity. These visual disturbances have been hypothesized to be related to increases in intracranial pressure. Increases in intracranial pressure associated with spaceflight are thought to occur through the cephalad shift of intravascular and cerebrospinal fluids due to the loss of the head-to-foot gravity vector present on Earth. In regard to the vascular component, increases in cerebral blood flow and capillary fluid filtration could elevate fluid movement into the cranium, while impaired function of cerebral veins and lymphatics could impede intracranial fluid drainage out of the cranium. Therefore, the purpose of this proposal is to examine the effects of long-term spaceflight on the vasoconstrictor, mechanical, and structural properties of cerebral arteries, veins, and lymphatic vessels. Flight: The aim of this project is to investigate whether spaceflight on the ISS alters the blood-brain barrier in rodents, as indicated by ultrastructural examination of the cerebral capillary endothelium. To assess the changes that occur in the structure of the blood-brain barrier, 10-15 female C57BL/6 mice, 10-11 weeks old will be loaded into a transporter (Animal Enclosure Module AEM-T; 10 mice/side), and late-loaded into a SpaceX Dragon rocket. After ascent and docking to the ISS, the mice will be transferred via an Animal Access Unit (AAU) and Mouse Transfer Box (MTB) to 2 Habitats (AEM-X). The mice will be grouped into four experimental groups, each housed in one side of an AEM-X. The groups, as determined by another investigation, will consist of mice receiving intraperitoneal (i.p.) injection of saline vehicle (n=5), Tetanus Toxoid (TT) vaccine (n=5), CpG ODN adjuvant (n=5), or TT vaccine + CpG ODN adjuvant (n=5). After mice have been exposed to microgravity for a minimum of 2 weeks, animals for this study will be inoculated with an injection of phosphate buffered saline (n=5) or CpG ODN adjuvant (n=5); a third group of animals injected with the TT vaccine (n=5) may be utilized. Two weeks later (14 +/- 2 days), the mice will be sacrificed in the Materials Science Glovebox (MSG). Sacrificing will be achieved via i.p. injection of 100/10 mg/kg ketamine/xylazine anesthesia followed by exsanguination by cardiac puncture. Following blood collection and harvest of spleen for another research investigation, the animals will have heads removed. The top of the skull will be cut along the sagittal crest and suture and retracted back to expose the brain. The brains will be will be dissected from the mice and put in a solution of 10% buffered formalin at ambient temperature. 24-48 hours later, the 10% buffered formalin will be exchanged with the Brain Fixation Kit (BFK) to 70% Ethanol and stored at ambient temperature until they can be returned to Earth and analyzed. Ground: 20-25 (saline vehicle (n=10), Tetanus Toxoid (TT) vaccine (n=5), CpG ODN adjuvant (n=10)) ground control mice will be put in AEMs inside Orbiter Environmental Simulators (OES) on a 2-8 day delay, depending on the when telemetry data from the SpaceX Dragon and ISS environments are available. The differences in vascular structure between ground controls and flight animals with a P-value < 0.05 will be considered statistically significant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The lymphatic system is one of the least understood parts of the circulatory system in the body. This project will be the first to examine the effects of microgravity on lymphatic function. On Earth, understanding gained from these studies will help us better understand the adaptability of the lymphatic circulation and how it might respond to chronic bedrest and physical deconditioning. These studies may also provide insight into intracranial hypertension that occurs on Earth.
Task Progress:	These studies will be conducted on mice who have flown to the International Space Station. The mice are scheduled to be launched on August 3, 2017, on the SpaceX-12 mission. Consequently, there are no results to be reported for this phase of the project To aid with the training of laboratory personnel in the conduct of microvessel studies specific to this grant, cerebral and coronary arteries were isolated from 1) control mice, 2) mice that had undergone 2 weeks of hindlimb unloading (HU), 3) mice exposed to total body irradiation TBI; 1 Gy of 56Fe ions, 600 MeV/nucleon, LET 150 keV/µm in water), and 4) mice undergoing both HU and TBI (HU+TBI) to determine the effects of these treatments on levels of the pro-oxidant proteins xanthine oxidase (XO) and NADPH oxidase-2 (NOX-2), anti-oxidant protein superoxide dismutase-1 (SOD-1), and endothelium-dependent vasodilator signaling protein endothelial NOS (eNOS). These studies with cerebral and coronary arteries were part of a bigger project funded by the National Space and Biomedical Research Institute and NASA Space Biology to determine the effects of these treatments simulating weightlessness and space relevant irradiation on peripheral vascular function of the gastrocnemius feed artery. Results obtained from the coronary artery (funded from the present NASA Space Biology grant) were published with results from the gastrocnemius feed artery (see Bibliography below). These results are significant in that they demonstrates that vascular adaptation to simulated weightlessness, irradiation, and the combination of the two can differ in vascular beds of different tissues, e.g., coronary vs. cerebral. If these findings occur in a true space environment, it suggests that countermeasures to maintain vascular health may have varying degrees of effectiveness depending on the tissue.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Articles in Peer-reviewed Journals	Delp MD, Charvat JM, Limoli CL, Globus RK, Ghosh P. "Apollo lunar astronauts show higher cardiovascular disease mortality: Possible deep space radiation effects on the vascular endothelium." Scientific Reports. 2016 Jul 28;6:29901. http://dx.doi.org/10.1038/srep29901 ; PubMed PMID: 27467019; PubMed Central PMCID: PMC4964660 , Jul-2016